

ETHAN SAVAR

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EDUCATION

The Ohio State University

Columbus, OH

B.S. in Computer Science Engineering, Mathematics Double Major

April 2026

- Integrated Business and Engineering Honors Program, Teaching Assistant

EXPERIENCE

JPMorganChase | Software Engineering Intern

June 2025 – Aug 2025

- Built a full-stack analysis tool for identifying recurring server-side failure signals for trading workflow applications, gathering data from multiple sources into a single dashboard, enabling a reduction in response to incidents by 18%
- Developed real-time APIs using Java and Spring Boot, and a React frontend displaying graphs, filtering, and search
- Created a configurable health-check and SQL probe runner, enabling teams to create/parameterize queries and validate service redundancy leading to a reduction in manual verification by 3-4 hours/week

Immuta | Research/Data Scientist Intern

May 2024 – May 2025

- Developed a regex to automata compiler in Python using NFA/DFA subset conversion and Hopcroft minimization, reducing sensitive data discovery detection error rate below 1% and improving coverage by 36% across data classes
- Reduced AI Copilot costs by 12% while doubling throughput by introducing RAG context caching and prompt compression in AWS Bedrock
- Created hierarchical semantic buckets for regex using BERT embeddings and agglomerative clustering in PyTorch
- Built a LLM prompt evaluation tool to quantify output drift across identical prompts through embedding cosine similarity and KS tests

Scarlet Investment Group | VP of Quantitative Development

Jan 2024 – Dec 2025

- Co-founded Ohio State University's first quantitative finance organization and defined the club's technical vision
- Led and managed 20+ engineers across multiple projects such as a back-tester, competitions, and writing a research paper on the mathematical validity of a novel trading strategy
- Partnered with industry leaders to design and compete in real-world trading challenges, bridging academic practices

The Ohio State University | Undergraduate Research Assistant

Oct 2023 – Dec 2024

- Applied stochastic differential equations to improve diffusion models for video generation motion consistency
- Designed an uncertainty calibration method for LLMs that transforms logit outputs to estimate model confidence
- Built a video analysis pipeline using Python and CLIP, enabling natural language search of video scenes

PROJECTS

AutoTex: Language to LaTeX Conversion | React, Tailwind, Electron, Ollama, Python, PyTorch

- Built a cross-platform desktop app that converts natural language to structured LaTeX using LLM-based parsing
- Integrated a lightweight CNN/OCR model in PyTorch to extract and interpret handwritten notes and PDFs, achieving ~94% character-level accuracy on custom datasets.

Custom Language Compiler | Java

- Designed and implemented a custom compiler, parser, and interpreter for a domain-specific programming language
- Built lexical analysis and recursive descent parsing with AST construction, supporting type inference, error handling, and modular execution.

Job Placement Locator | Python, Tableau

- Collaborated and won (~20 teams) at JPMC's Data For Good Hackathon by designing a Tableau dashboard and Python-based ranking algorithm to identify U.S. regions for nonprofit intervention targeting disadvantaged students
- Developed a composite metric leveraging job market density, GDP, university distribution, and population demographics, enabling data-driven prioritization of outreach areas.

TECHNICAL SKILLS

- **Languages:** Java, Python, TypeScript, JavaScript, C, SQL, HTML, CSS, x86 Assembly
- **Technologies:** PyTorch, AWS, Docker, React, PostgreSQL, Spring Boot, Maven, Git